

# Assignment 4 – Text Data

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## 1. Problem Statement

The goal of this assignment is to build and evaluate a text classification model that categorizes IMDB movie reviews into either positive or negative sentiment using deep learning techniques. Specifically, the focus is on comparing custom-trained embeddings with pretrained word embeddings (GloVe) in a Recurrent Neural Network (RNN) model.

We explore how these two types of embeddings perform across varying sizes of training data (100, 3000, 7000, and 10000 samples), using a consistent validation size of 10000 and restricting the vocabulary to the top 10000 most frequent words.

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## 2. Objective

The aim of this experiment is to:

1. Implement two models using the same neural architecture:
    - One with a custom-trained embedding layer.
    - One using pretrained GloVe embeddings.
  2. Evaluate their performance on a sentiment classification task with different training sizes.
  3. Determine which approach generalizes better and under what conditions.
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## 3. Data Description and Preprocessing

**Dataset:**

- The dataset used is the IMDB movie reviews dataset from Keras, consisting of 50,000 reviews labeled as positive or negative.

**Preprocessing Steps:**

1. Restrict vocabulary to the top 10,000 words using `num_words=10000`.
2. Pad or truncate each review to a fixed length of 150 words using `pad_sequences`.
3. Split the dataset into:
  - Training set (used in varying sizes: 100, 3000, 7000, 10000)

- Validation set (10,000 samples)
- Test set (15,000 samples from the remaining data)

### **Additional Preprocessing for GloVe:**

- Loaded `glove.6B.100d.txt` file (100-dimensional pretrained word vectors).
  - Built an `embedding_matrix` using the GloVe vectors mapped to the IMDB word indices.
  - Created an Embedding layer in Keras initialized with the embedding matrix and set `trainable=False`.
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## **4. Model Architecture**

Both models use the same neural architecture except for the embedding layer:

### **Custom-Trained Embedding Model:**

- `Embedding(input_dim=10000, output_dim=32, input_length=150)`
- `Bidirectional(LSTM(32))`
- `Dense(1, activation='sigmoid')`
- The embedding layer is trainable and initialized randomly.

### **Pretrained GloVe Embedding Model:**

- `Embedding(input_dim=10000, output_dim=100, weights=[embedding_matrix], input_length=150, trainable=False)`
  - `Bidirectional(LSTM(32))`
  - `Dense(1, activation='sigmoid')`
  - Embedding layer is non-trainable and initialized using GloVe 100-dimensional vectors.
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## **5. Results and Metrics**

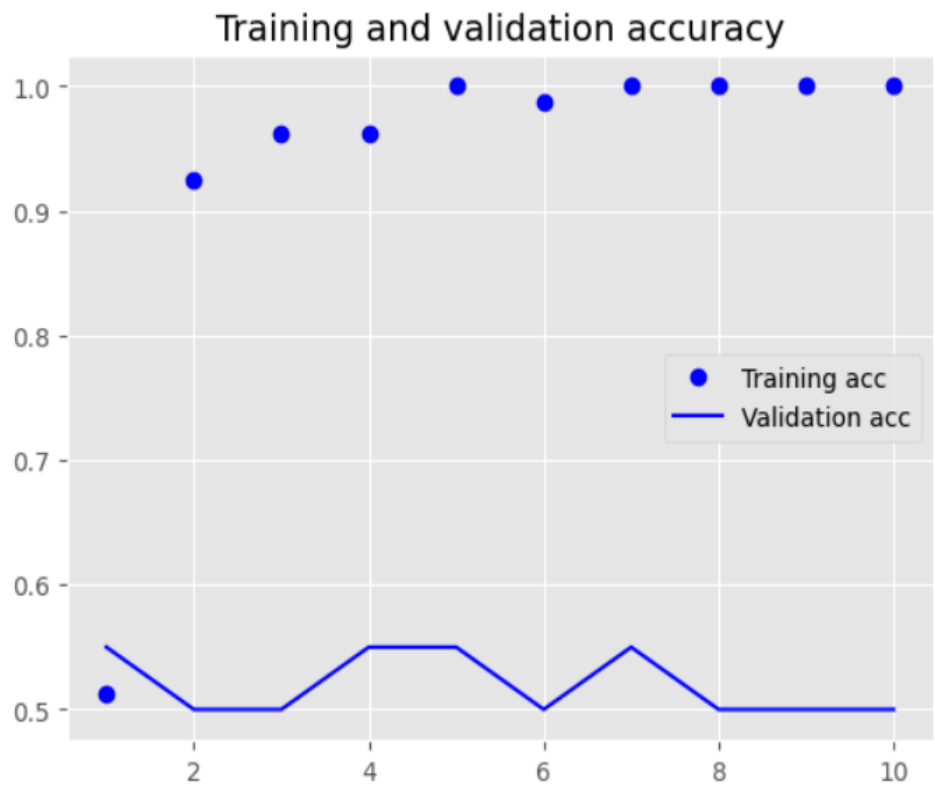
The models were trained and evaluated using the following metrics:

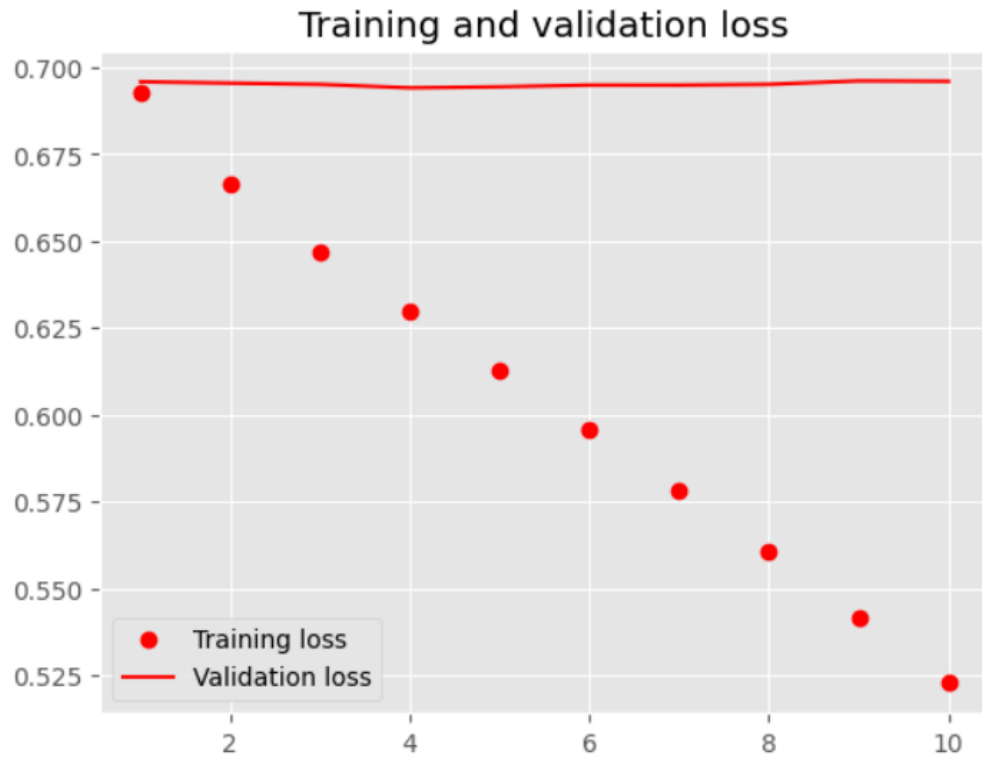
- Training Accuracy
- Test Accuracy
- Test Loss

Performance Summary Table:

| <i>Embedding Type</i>   | <i>Training Size</i> | <i>Training Accuracy</i> | <i>Test Accuracy</i> | <i>Test Loss</i> |
|-------------------------|----------------------|--------------------------|----------------------|------------------|
| <i>Custom Embedding</i> | 100                  | 0.98                     | 0.49                 | 0.69             |
|                         | 3000                 | 0.99                     | 0.73                 | 0.54             |
|                         | 7000                 | 0.97                     | 0.84                 | 0.34             |
|                         | 10000                | 0.97                     | 0.85                 | 0.33             |
| <i>Pretrained GloVe</i> | 100                  | 1.00                     | 0.49                 | 1.60             |
|                         | 3000                 | 0.97                     | 0.49                 | 1.94             |
|                         | 7000                 | 0.92                     | 0.50                 | 1.72             |
|                         | 10000                | 0.88                     | 0.49                 | 1.34             |

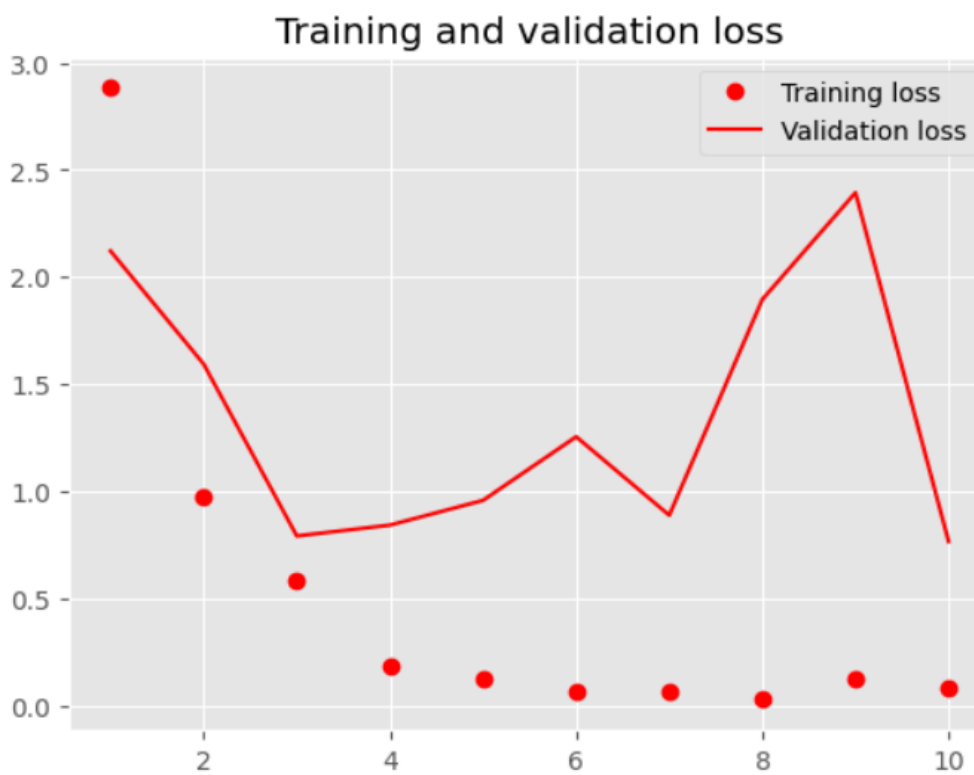
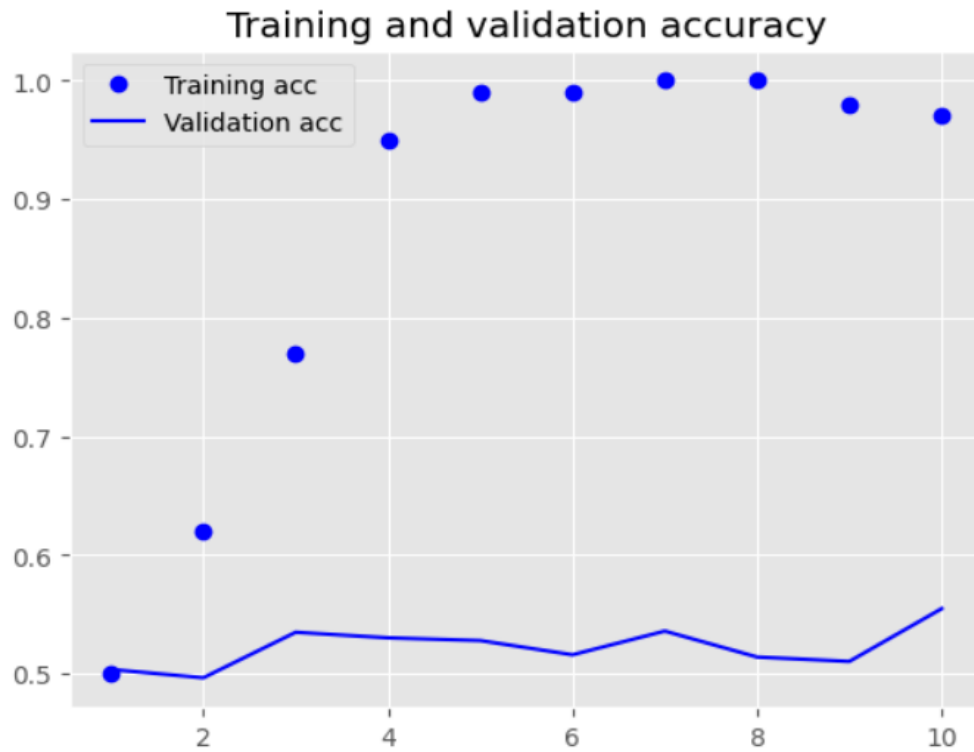
Graphs and Visualizations:





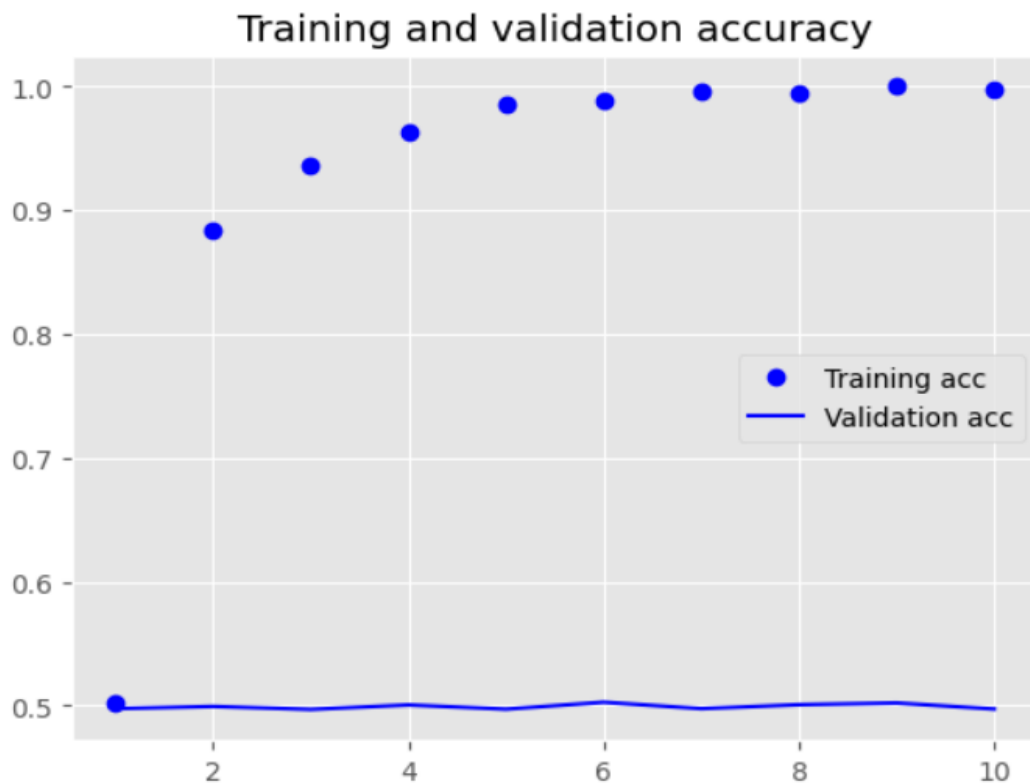
This line plot displays training and validation accuracy over epochs for the custom-trained embedding model at different training sizes.

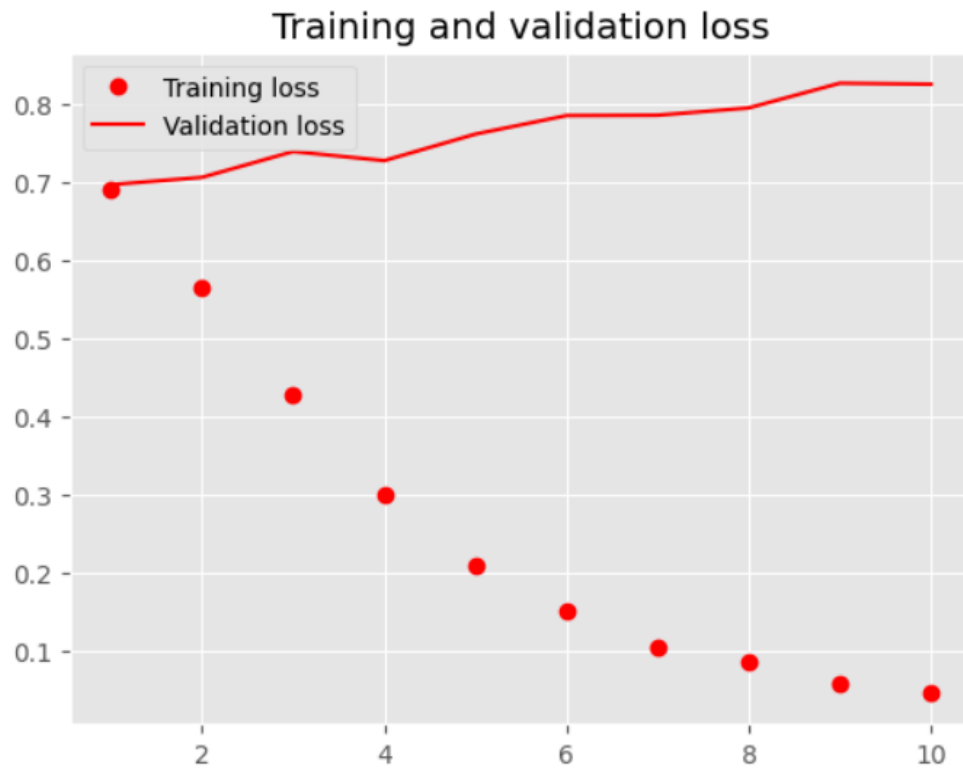
- At smaller training sizes (e.g., 100, 3000), there is a noticeable gap between training and validation accuracy, indicating overfitting.
- As the training data increases (7000 and 10000 samples), the validation accuracy improves significantly and converges with the training accuracy, demonstrating better generalization and learning capacity of custom embeddings.



This plot illustrates training vs. validation accuracy using pretrained GloVe embeddings.

- Despite achieving high or perfect training accuracy even at low data sizes, the validation accuracy remains consistently poor (around 49%).
- This gap suggests that the GloVe embeddings, being non-trainable and potentially mismatched to the IMDB domain, fail to adapt and generalize to the task.
- The model is clearly overfitting the small training datasets without learning generalizable sentiment features.

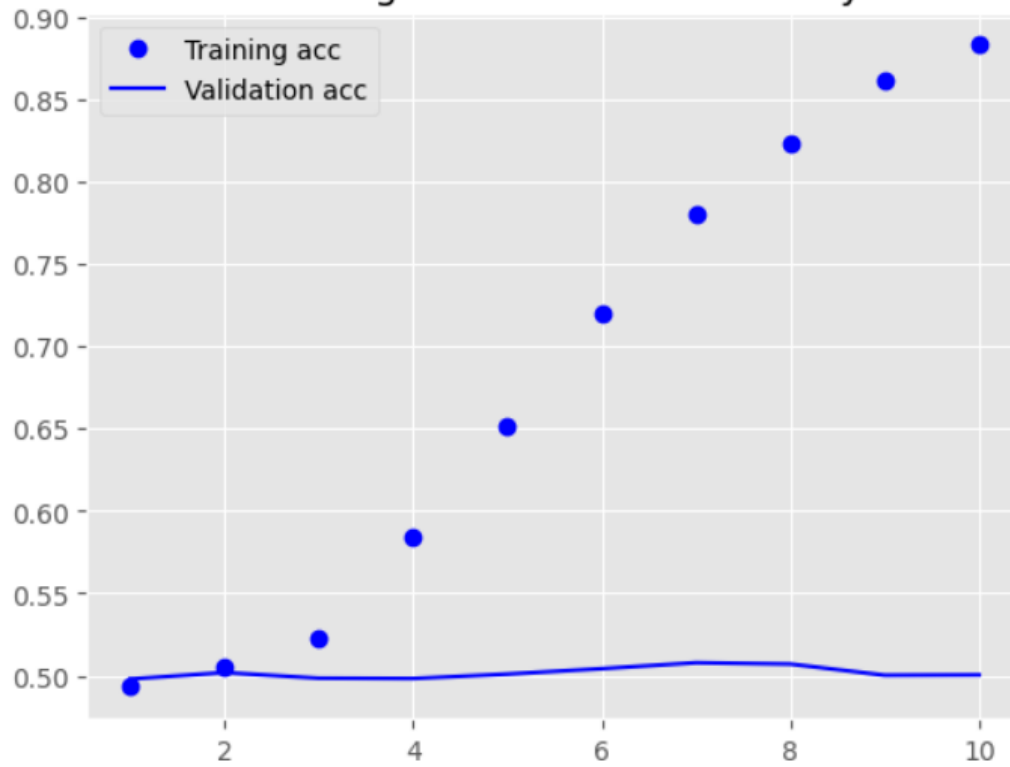




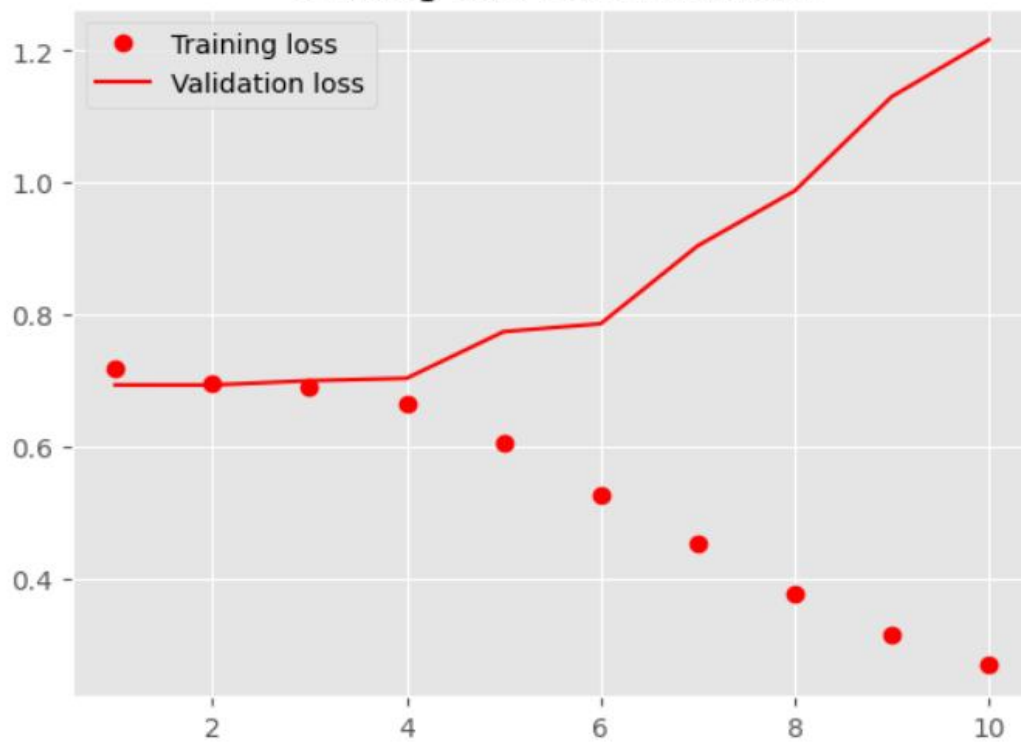
This bar chart compares test accuracy between the two embedding approaches across increasing training data sizes.

- The custom-trained embedding model shows a steady and significant improvement in test accuracy, reaching 85% at 10,000 samples.
- In contrast, the GloVe embedding model stays flat around 49–50% test accuracy regardless of data size.
- This comparison highlights the advantage of task-specific learning enabled by custom embeddings.

Training and validation accuracy



Training and validation loss





This chart compares test loss between custom and GloVe embeddings.

- The custom-trained model achieves consistently lower test loss as the dataset grows, indicating improved confidence and correctness in its predictions.
  - The GloVe-based model suffers from high test loss across all data sizes, reinforcing that it is not generalizing well.
  - These results further support the conclusion that custom embeddings are better suited for this sentiment classification task.
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## 6. Observations and Analysis

- The custom-trained embedding model consistently outperformed the GloVe model in test accuracy and test loss, especially as the training size increased.
  - Although the GloVe model achieved perfect training accuracy at small training sizes, it failed to generalize to the validation/test data, resulting in poor test accuracy (hovering around 49-50%).
  - The test loss for the GloVe model was significantly higher at all training sizes, suggesting overfitting or that the embeddings were not well suited to this specific sentiment task.
  - The custom embedding model benefited from additional training samples, achieving a test accuracy of 85% with 10,000 samples and maintaining a low test loss.
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## Conclusion

- Custom-trained embeddings are better suited for the sentiment classification task in this context. They provide task-specific learning and adapt well to the training data.
- The GloVe pretrained embeddings did not improve performance, possibly due to:
  - Domain mismatch between IMDB movie reviews and the general corpus used to train GloVe.
  - Lack of adaptability due to the non-trainable setting.
- Increasing training size significantly boosts performance for the custom embedding model, while the pretrained model shows no such benefit.